Tabular Q-Learning on FrozenLake 8x8

CPSC 4420 Assignment 2 | Zach Fogle

This project implements tabular Q-learning, establishes a baseline, isolates the effects of learning rate, discount factor, and exploration rate, and evaluates the strongest combined policy without exploration.

1. Environment and method

Environment. Gymnasium FrozenLake-v1 uses the standard 8x8 map: 64 discrete position states and four actions (left, down, right, up). The goal is to move from the upper-left start to the lower-right goal without entering a hole. With **is_slippery=False**, transitions are deterministic. Reaching the goal gives reward 1; all other transitions give 0. An episode ends at a hole or goal, or is truncated at the time limit.

Agent. A zero-filled 64x4 Q-table is learned with epsilon-greedy action selection. With probability epsilon the agent explores randomly; otherwise it chooses a maximum-Q action, breaking ties randomly. Terminal transitions use no future-value term.

```
Q(s,a) <- Q(s,a) + alpha [r + gamma max_a' Q(s',a') - Q(s,a)]
```

2. Experimental setup and baseline

Map / dynamics	Episodes	Runs	Rolling window	Final window	Seed
8x8 / deterministic	10,000	10 per condition	200 episodes	Last 500	123 + run

The baseline used **alpha=0.50, gamma=0.90, epsilon=0.10**. Its rolling success rate rose sharply around episodes 1,000-2,000 and stabilized near 0.88 while constant exploration remained active.

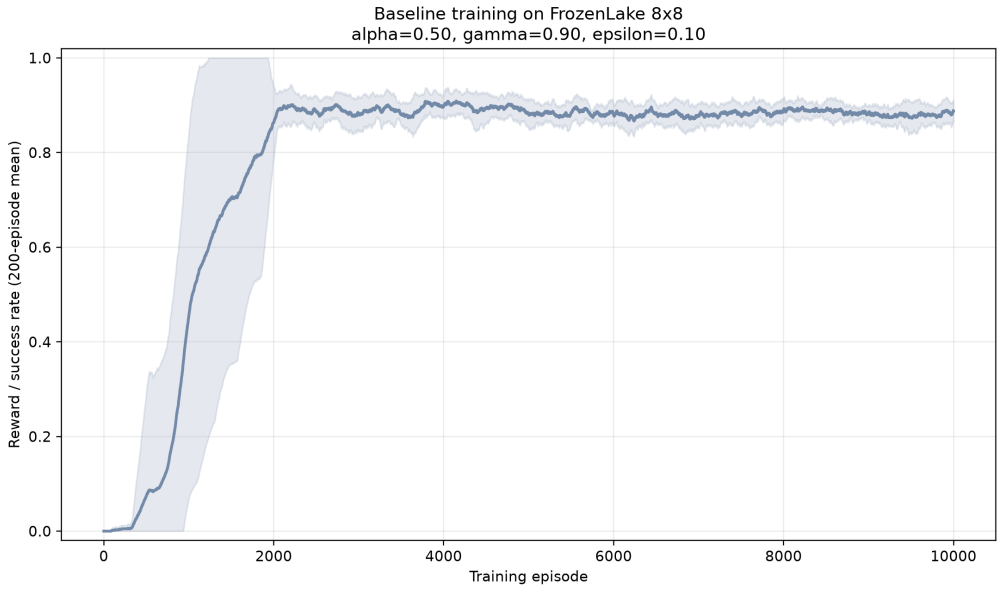
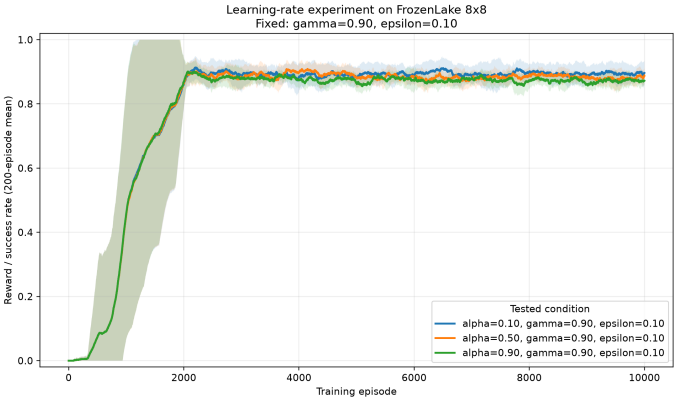


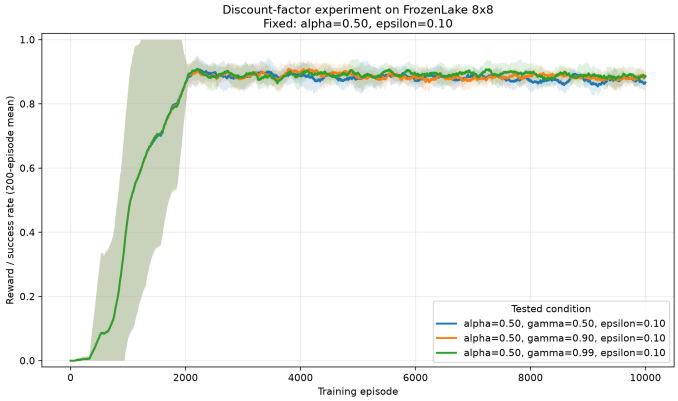
Figure 1. Baseline reward versus training episode; shading is variation across 10 runs.

3. Hyperparameter experiments

Only one parameter changes in each comparison. Curves show mean 200-episode rolling reward and shading shows one standard deviation across 10 runs.



Learning rate. alpha=0.10, 0.50, 0.90; gamma=0.90 and epsilon=0.10 fixed.



Discount factor. gamma=0.50, 0.90, 0.99; alpha=0.50 and epsilon=0.10 fixed.

Results. Learning speed was nearly identical: all alpha settings reached 0.50 success around episodes 1,036-1,044. alpha=0.10 had the highest final mean reward (0.895); alpha=0.90 had the lowest across-run standard deviation (0.011). Gamma settings also crossed 0.50 near episode 1,043. gamma=0.99 performed best (0.888), slightly ahead of 0.90 (0.884) and 0.50 (0.874), because it preserved the sparse future goal reward more strongly.

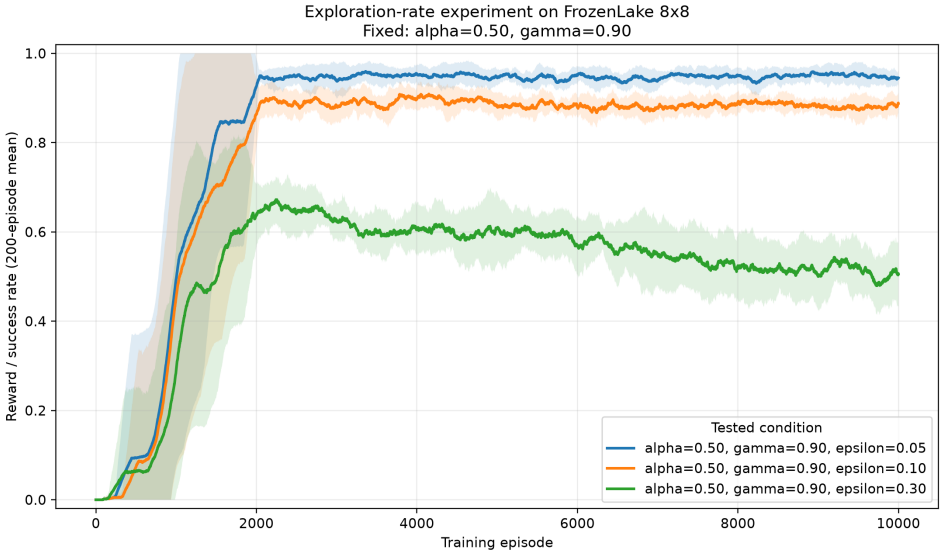


Figure 4. Exploration-rate comparison with alpha=0.50 and gamma=0.90 fixed.

Exploration rate had the largest effect. epsilon=0.05 reached 0.50 success first (episode 1,005) and achieved final mean reward 0.947. epsilon=0.30 learned more slowly and finished at 0.497 because 30% of its actions remained random even after it learned a successful route.

4. Best-agent evaluation and conclusions

The strongest settings were combined as **alpha=0.10, gamma=0.99, epsilon=0.05**. The selected run achieved final 500-episode training reward 0.974, versus 0.918 for the selected baseline run.

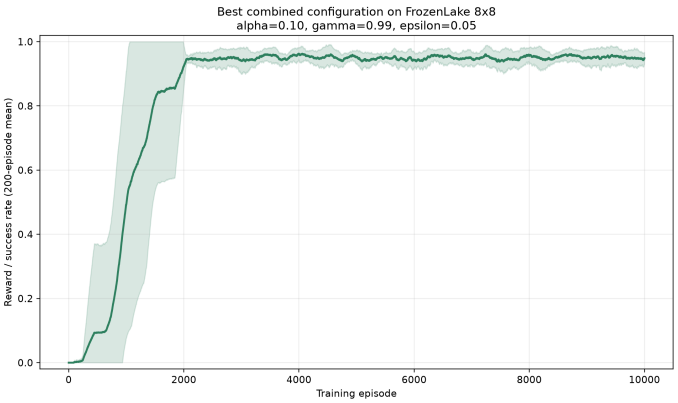


Figure 5. Combined-best training performance.

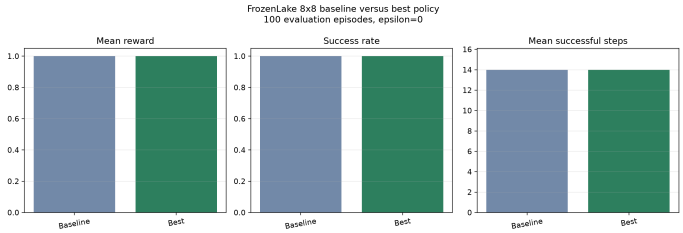


Figure 6. Greedy baseline-versus-best evaluation.

Agent	alpha	gamma	train epsilon	Mean reward	Success	Mean steps
Baseline	0.50	0.90	0.10	1.000	100%	14.0
Best	0.10	0.99	0.05	1.000	100%	14.0

Each policy was evaluated for 100 episodes with epsilon=0 and no Q-table updates. Both succeeded every time in 14 steps, so both learned the same shortest route. The best configuration's advantage appears during training: lower exploration causes fewer random failures after convergence.

Conclusion

This deterministic task was relatively insensitive to the tested alpha and gamma values after sufficient training. Constant exploration mattered most: a low nonzero epsilon discovered the route while avoiding excessive random failures. Reporting both training reward and greedy evaluation is important, because exploration lowers training reward even when the learned policy is fully reliable.

References

Farama Foundation, *Solving FrozenLake with Tabular Q-Learning*, gymnasium.farama.org/tutorials/training_agents/frozenlake_q_learning/. Sutton and Barto, *Reinforcement Learning: An Introduction*, 2nd ed., MIT Press, 2018.